Basic Information

Experiment No.: 09

Date: 11-10-2026

Title: Creating and Importing a Package to Convert Roman Numerals into Integers

---

Objective / Aim

To create a user-defined package named "number", define a "Roman" class with the "romanToInteger()" method, and import and use the method in another Java class.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
- Any Java IDE or Command Prompt
- Operating System: Windows/Linux

Algorithm / Logic Description

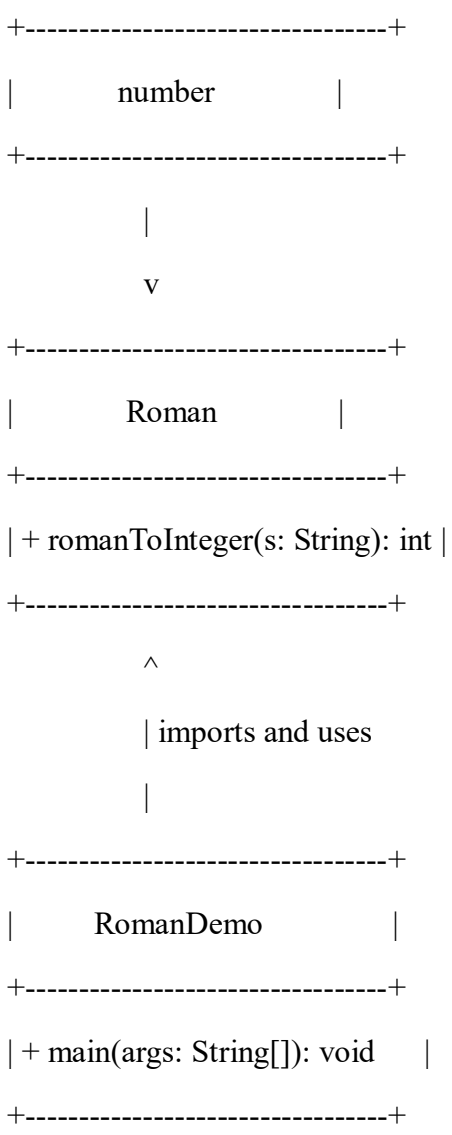
1. Create a package named "number".
2. Define a class named "Roman".
3. Create the "romanToInteger()" method to convert Roman numerals into integers.
4. Assign integer values to Roman symbols.
5. Subtract a symbol's value if it is smaller than the next symbol; otherwise, add it.

6. Return the calculated integer.

7. Import the "Roman" class into another class.

8. Read a Roman numeral and display its integer value.

📐 UML Class Diagram



💻 Source Code

File 1: Roman.java

```
package number;
import java.util.HashMap;
import java.util.Map;

public class Roman {

    // Converts a Roman numeral into an integer
    public int romanToInteger(String s) {

        // Store the values of Roman symbols
        Map<Character, Integer> values = new HashMap<>();

        values.put('I', 1);
        values.put('V', 5);
        values.put('X', 10);
        values.put('L', 50);
        values.put('C', 100);
        values.put('D', 500);
        values.put('M', 1000);

        int result = 0;

        // Process each character in the Roman numeral
        for (int i = 0; i < s.length(); i++) {

            int current = values.get(s.charAt(i));
```

```
            // Subtract if the current value is smaller
            // than the next value
            if (i + 1 < s.length()
                    && current < values.get(s.charAt(i + 1))) {
                result -= current;
            } else {
                // Otherwise, add the current value
                result += current;
            }
        }

        return result;
    }
}
```

File 2: RomanDemo.java

```
import java.util.Scanner;
import number.Roman;

public class RomanDemo {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Create an object of the Roman class
        Roman r = new Roman();
```

```
        // Read the Roman numeral
        System.out.print("Enter a Roman numeral: ");
        String s = sc.nextLine().toUpperCase();

        // Convert and display the result
        int result = r.romanToInteger(s);

        System.out.println("Integer value: " + result);

        sc.close();
    }
}
```

🔨 Compilation & Execution Commands

Folder Structure

```
Project/
│
├── number/
│   └── Roman.java
│
└── RomanDemo.java
```

Open the terminal in the "Project" folder.

Compile

```
javac number/Roman.java RomanDemo.java
```

Run

```
java RomanDemo
```

Output Screen

Enter a Roman numeral: LVIII

Integer value: 58

Explanation: L = 50, V = 5, III = 3. Therefore, LVIII = 50 + 5 + 3 = 58.

Result

The user-defined package "number" was successfully created. The "Roman" class and its "romanToInteger()" method were implemented and imported into another class to convert Roman numerals into integers.